

Solution (sketch) to IMO 2025 Shortlist

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Algebra

- A1. Quadratic solitaire is a single-player game. To start the game, the player chooses two distinct nonzero integers a and b and writes the equation $x^2 + ax + b = 0$ on a blackboard. On each turn, if the equation currently written on the blackboard has two distinct nonzero integer solutions $x = u$ and $x = v$, then the player erases the equation and replaces it by one of the equations $x^2 + ux + v = 0$ and $x^2 + vx + u = 0$ of their choosing. Otherwise, the game ends. Determine all initial choices of a and b such that quadratic solitaire can be played forever.

Answer. $(1, -2)$ where $x^2 + x - 2 = (x - 1)(x + 2)$ gives the same polynomial.

Solution. Now note that $uv = b$ so $|b| \geq \max\{|u|, |v|\}$ (as each $|u|, |v| \geq 1$) so $|b|$ is nonincreasing, hence needs to stabilize. Note that $|v| = |b|$ means $|u| = 1$, hence eventually we end up with $x^2 \pm x + v$ with one of $\{1, -1\}$ as root. Since $v \neq 0$, $x^2 \pm x \in \{-2, 0, 2\}$ when $x = \pm 1$, so $|v| = 2$. $x^2 \pm x + 2$ has no real root as discriminant is -7 , hence $v = -2$. Now $x^2 - x - 2 = (x + 1)(x - 2)$ has root $\{-1, 2\}$, but arrangement into $x^2 - x + 2$ has no real root. Thus the only ‘terminal’ polynomial is $x^2 + x - 2$.

- A2. The sunshine cost of a sequence $a_1, a_2, \dots, a_{100}$ of integers is the largest possible value of

$$|(a_1 + a_2 + \dots + a_i) - a_j|$$

as i and j vary over all integers $1, 2, \dots, 100$.

Determine the smallest possible sunshine cost over all sequences $a_1, a_2, \dots, a_{100}$ of pairwise distinct integers.

Answer. 75, or $\lceil \frac{3(n-1)}{4} \rceil$ by replacing 100 with general n that’s multiple of 4.

Solution. *Lower bound:* we can’t choose a_1 as the maximizer of $|a_i|$; otherwise either a_1 is the max or min and $|a_1 - a_i| \geq n - 1$ for some i .

Let $\max a_j - a_i = M$ satisfied by (j_0, i_0) and $\max |a_i| = C$; w.l.o.g. let $a_k = C$, $k \geq 2$. clearly can’t have $|a_1| = C$. Let the sum $(a_1 + \dots + a_i) - a_j = C_{i,j}$. Now

$$C_{k,i} - C_{k-1,j} = a_k + a_j - a_i = C + M$$

so the answer is at least $\frac{C+M}{2}$. Moreover $M \geq n - 1$, $C \geq \lceil \frac{n-1}{2} \rceil$, giving the LB.

Upper bound: we’ll only show the case $n = 4k$ as requested in the problem, and the desired bound is $3k$. We’ll make sure all $|a_i| \leq 2k$; all $|a_1 + \dots + a_i| \leq k$. We arrange as

$$-k, 2k, -2k, 0, 1, 2k - 1, -1, -(2k - 1), \dots, k - 1, k + 1, -(k - 1), -(k + 1)$$

- A3. (IMO 5) Alice and Bazza are playing the inekoalaty game, a two-player game whose rules depend on a positive real number λ which is known to both players. On the n th turn

of the game (starting with $n = 1$) the following happens: If n is odd, Alice chooses a nonnegative real number x_n such that

$$x_1 + x_2 + \cdots + x_n \leq \lambda n.$$

If n is even, Bazza chooses a nonnegative real number x_n such that

$$x_1^2 + x_2^2 + \cdots + x_n^2 \leq n.$$

If a player cannot choose a suitable x_n , the game ends and the other player wins. If the game goes on forever, neither player wins. All chosen numbers are known to both players. Determine all values of λ for which Alice has a winning strategy and all those for which Bazza has a winning strategy.

Answer. A wins when $\lambda > \frac{\sqrt{2}}{2}$; B wins when $\lambda < \frac{\sqrt{2}}{2}$.

Solution. Let $C = \frac{\sqrt{2}}{2}$.

$\lambda > C$: Coral outputs 0 at all times; Joey's outputs will make sure $\sum x_n \leq Cn$. Wait till n big enough (and even) then pick $x_{n+1} = (\lambda - C)n$ (then $\sum x_n^2 \geq (\lambda - C)n^2 > n$), Joey loses.

$\lambda < C$: Joey always tries to 'max out', i.e. $x_{2k+1}^2 + x_{2k+2}^2 = 2$. Note that $x_{2k+1} + x_{2k+2} \geq \sqrt{2}$ so Coral loses at one point.

$\lambda = C$: Coral sustains by outputting 0; Joey sustains by making sure $x_{2k+1}^2 + x_{2k+2}^2 = 2$.

- A4. Let $\mathbb{Z}_{\geq 0}$ be the set of all nonnegative integers. Let $f : \mathbb{Z}_{\geq 0} \rightarrow \mathbb{Z}_{\geq 0}$ be an unbounded function such that, if m and n are nonnegative integers satisfying

$$f(m+n) = \max\{f(0), f(1), \dots, f(m+n)\},$$

then

$$f(m+n) = f(m) + f(n).$$

Prove that there exist positive integers A, B, C and D such that for all nonnegative integers n ,

$$f(An + B) = Cn + D.$$

Solution. Denote peak as $k \geq 1$ satisfying $f(k) \geq f(\ell)$ for all $\ell \leq k$. Let $C = \min\{f(m) : m \geq 1\}$. There are infinitely many peaks since f is unbounded. We show the following:

$C > 0$. Indeed, if $f(m) = 0$ for $m \geq 1$ then for any peak $n \geq m$ we have $f(n) = f(n-m)$ and so f is bounded.

Exactly one m has $f(m) = c$. Let $m_1 < m_2$ be such that $f(m_1) = f(m_2) = C$. For any peak $n > m_2$, we have $f(n) = f(n-m_1) + C = f(n-m_2) + C$ and note that both $f(n-m_1), f(n-m_2)$ are $\max\{f(0), \dots, f(n-1)\}$. Since $m_1 < m_2$ we have $f(n-m_1) = f(n-m_2) + f(m_1-m_2)$ and $f(m_1-m_2) = 0 < C$, which is contradiction.

If n is a peak, $f(A) = C$, and $n > A$, then $n-A$ is a peak. For any $m < n$ we have $f(n-A) = f(n) - f(A) \geq f(n) - f(m) = f(n-m)$ so $f(n-A) = \max\{f(0), \dots, f(n-1)\}$.

The last point says the number of peaks among $\{kA+1, \dots, (k+1)A\}$ is at most that of $\{(k-1)A+1, \dots, kA\}$; choose such k such that this number stabilizes and let $B \in \{kA+1, \dots, (k+1)A\}$ be a peak. Then $B, A+B, 2A+B, \dots$ are all peaks. Therefore $f(nA+B) = nf(A) + f(B) = Cn + D$, as claimed.

- A6. Let S be a set of positive integers, possibly infinite, such that no positive integer greater than 1 divides all elements of S . Determine all non-periodic infinite sequences $a_1, a_2, a_3, \dots$ of positive integers such that, for all positive integers n ,

- $a_n \leq |a_{n+\ell} - \ell|$ for all ℓ in S , and
- $a_n = |a_{n+\ell} - \ell|$ for at least one ℓ in S .

Answer. $a_n = n + c$ for $c \geq 0$, which obviously fulfills.

Solution. We rewrite the condition in the following form:

$$a_n = \min\{|a_{n+\ell} - \ell| : \ell \in S\} \cdots (*)$$

Lemma 1. *All bounded sequences in $(*)$ are periodic.*

Proof. Let M be $\max\{a_n\}$, then $M < |a_{n+\ell} - \ell|$ whenever $\ell > 2M$. Denote $s_{n,k} = (a_{n+1}, \dots, a_{n+k})$; a_n is a function of $s_{n,2M}$. Let h_n be the minimum positive integer such that $s_{n,2M} = s_{n-h_n,2M}$ (or ∞ if this number doesn't exist), then $s_{n,2M} = s_{n-h_n,2M} \rightarrow a_n = a_{n-2M} \rightarrow s_{n-1,2M} = s_{n-h_n-1,2M}$ so $h_{n-1} \leq h_n$ and given that there are at most M^{2M} distinct possibilities of s_n , $h_n \leq M^{2M}$ for infinitely many n 's, and therefore $h_n \leq M^{2M}$ for all n . Choose $h = \max\{h_n\}$, we have a_n periodic with period h . $\square$

Now among unbounded sequences, it suffices to show that $a_n = n + c$ for n sufficiently large by $(*)$. Denote $b_n = a_n - n$ and $m = \lim_{n \rightarrow \infty} \inf\{b_n\}$. ; the second bullet point $a_n = |a_{n+\ell} - \ell|$ means either $b_{n+\ell} = b_n$ or $b_{n+\ell} = -a_n - n < b_n$; so either way $b_{n+\ell} \leq b_n$ and iteratively, we get $m \leq b_n$ for all n . On the other hand, choose a finite set S_1 such that $\gcd S_1 = 1$, n_0 such that $a_{n_0} > \max S_1$. Then $a_{n_0+\ell} \geq a_{n_0} + \ell$ for all $\ell \in S_1$ (i.e. $b_{n_0+\ell} \geq b_{n_0}$). Let p_0 be such that for all integers $p \geq p_0$, p can be written as nonnegative integer combinations of S_1 (which is valid as $\gcd S_1 = 1$, can similarly deduce $b_{n_0+p} \geq b_{n_0}$ for all $p \geq p_0$, and since $a_{n_0+p} > \max S_1$, this also gives $b_{n_1+p} \geq b_{n_1}$ for all $n_1 > n_0 + p$ (i.e. $m \geq b_n$ for all such n 's). Thus combining the arguments we have $m = b_n$ for all $n > n_1$, i.e. $a_n = n + m$ for all $n > n_1$, and so the argument follows.

- A7. Let $k \geq 3$ be a positive integer. Prove that there exists a unique k -tuple $(x_1, x_2, \dots, x_k)$ of positive real numbers such that $x_1 \geq x_2 \geq \dots \geq x_k$, and

$$\left\lfloor nx_1 + \frac{1}{2} \right\rfloor + \left\lfloor nx_2 + \frac{1}{2} \right\rfloor + \dots + \left\lfloor nx_k + \frac{1}{2} \right\rfloor = n$$

for every integer n .

Answer. $x_m = \frac{2^{k-m}}{2^k - 1}$.

Solution. Call all k -tuples working if it satisfies the problem condition. We start with the following set up:

- Picking n large and using $x - 1 < \lfloor x \rfloor \leq x$, we have $\sum x_i = 1$;
- $\lfloor y + \frac{1}{2} \rfloor + \lfloor -y + \frac{1}{2} \rfloor$ is 1 if $y + \frac{1}{2}$ is integer and 0 otherwise. Considering n and $-n$ simultaneously for each $n > 0$, nx_m cannot be half-integer.
- Let S_x be integers n where $\lfloor nx + \frac{1}{2} \rfloor - \lfloor (n-1)x + \frac{1}{2} \rfloor = 1$; $S_{x_1}, \dots, S_{x_k}$ are pairwise disjoint but union is $\mathbb{N}$.
- Any $x_1, x_2 \geq 0$ with sum 1 and nx_1 not a half-integer is a working pair.
- Denote $T_x = \min S_x \cap \mathbb{N}$; then $T_{x_1} = 1$. In addition, given that x_1 is such that nx_1 is not a half-integer, $S_{1-x_1} = \bigcup_{m=2}^k S_{x_m}$ and in particular, $T_{x_2} = T_{1-x_1} \geq 2$.
- Denote G_x the set of possible gaps between consecutive elements in S_x . We have $G_x = \mathbb{N} \cap [\lfloor \frac{1}{x} \rfloor, \lceil \frac{1}{x} \rceil]$. In addition, when $x < \frac{1}{2}$, we have $\frac{1}{2(t-1)} \leq x \leq \frac{1}{2t}$ where $t = T_x$, so $G_x \in \{2t-2, 2t-1, 2t\}$

We now consider the following steps.

Lemma 2. $T_{x_2} = T_{1-x_1} = 2$, $G_{x_2} = \{3, 4\}$ and $G_{x_1} = \{2, 3\}$.

Proof. Denote $T = T_{x_2}$ for simplicity. By inspecting any $t \in S_{x_3} > T_{x_2}$, and t_0, t_1 be consecutive elements in S_{x_2} such that $t_0 < t < t_1$, we have $2T \geq \max G_{x_2} \geq t_1 - t_0 \geq 2 \min G_{1-x_1} \geq 2(2T - 2)$, so $T \geq 2T - 2$ and $T = 2$. In addition, these inequalities must all hold so $\max G_{x_2} = 4, \min G_{1-x_1} = 2$. Finally, both of them must include 3, otherwise either $x_2 = \frac{1}{4}$ or $1 - x_1 = \frac{1}{2}$. These cases mean $2x_2$ and x_1 are half-integer, which is a contradiction. $\square$

Next, we note that each S_x partitions the real number line into intervals. We now consider the ‘blocks’ of intervals as follows: m blocks are defined by $m + 1$ points including both endpoints.

Lemma 3. All consecutive blocks of interval length 2 in S_{1-x_1} must contain even and equal number of intervals.

Proof. If $t \in S_{1-x_1}$ is an endpoint to an interval of length 3, and $t \notin S_{x_2}$, then S_{x_2} will have a gap of at least $2 + 3 \geq 5$ which contradicts $G_{x_2} = \{3, 4\}$. Conversely, for each of the intervals of size 2 in S_{1-x_1} , one of them cannot belong to x_2 . This leaves each consecutive blocks to have even number of 2’s, with the alternating ones (including both endpoints) belong to S_{x_2} and the other alternating ones $\cup_{i=3}^k S_{x_i}$.

Now, if $a \geq 2$ is the number of consecutive intervals of size 2 in S_x , i.e. for some t , $t, t+2, \dots, t+2a \in S_x$, but $t-a, t+2a+2 \notin S_x$. Then

$$\lfloor (t+2a)x + \frac{1}{2} \rfloor - \lfloor (t-1)x + \frac{1}{2} \rfloor \geq a+1 \quad \lfloor (t+2a+2)x + \frac{1}{2} \rfloor - \lfloor (t-3)x + \frac{1}{2} \rfloor \geq a+1$$

The first implies $(2a+1)x > a$ and $x > \frac{a}{2a+1}$; the second implies $(2a+5)x < a+2$ so $x < \frac{a+2}{2a+5}$. Thus there cannot be intervals of a blocks and b blocks simultaneously if $b \geq a+2$, proving the claim. $\square$

Consequently, if $2a$ is the consecutive block size (of interval 2) then we in fact have $1 - x_1 = \frac{2a+1}{4a+3}$, with

$$S_{1-x_1} = \{m(4a+3)+2p, p = 1, 2, \dots, (2a+1)\} \quad S_{x_2} = \{m(4a+3)+4p+2, p = 0, 1, \dots, a\}$$

For convenience, denote $M = 4a + 3$. Our goal is to show that $M = 2^m - 1$. We now iteratively establish the following about S_{x_m} for $m = 3, \dots, k$.

Lemma 4. For all m with $m \leq k$, $U_m := \cup_{m'=m}^k S_{x_{m'}} = \{Mu + 2^{m-1}c, c = 1, 2, \dots, \lfloor \frac{M}{2^{m-1}} \rfloor\}$, and in addition $U_m \cap \{1, 2, \dots, M\}$ has odd number of elements (so $2^m \mid M+1$).

Proof. We proceed by induction on m . For $m = 3$ the previous discussion already gave us $U_3 = \{4, 8, \dots, M-3\} \pmod{M}$, and $4 \mid M+1$. Our discussion will first assume this form of U_m and that $2^{m-1} \mid M+1$, show that U_m has odd number of elements, $2^m \mid M+1$, and if $m < k$ the next set U_{m+1} will fulfill the first condition.

Now assume $U_m \cap \{1, 2, \dots, M\} = \{2^{m-1}c, c = 1, 2, \dots, \lfloor \frac{M}{2^{m-1}} \rfloor\}$, in particular $T_{x_m} = 2^{m-1}$. Since $G_{x_m} \in \{2^m - 2, 2^m - 1, 2^m\}$, this would give $S_{x_m} \cap \{1, 2, \dots, M\} = \{2^{m-1} + 2^m c, c = 0, 2, \dots, \lfloor \frac{M-2^{m-1}}{2^m} \rfloor\}$. But the next element of U_m after M is $M + 2^{m-1}$, showing that

$$M + 2^{m-1} - 2^m \lfloor \frac{M - 2^{m-1}}{2^m} \rfloor \leq \max G_{x_m} \leq 2^m$$

So the remainder of $M - 2^{m-1}$ when divided by 2^m must be less than 2^{m-1} , which, combined with previous evidence, gives $M \equiv 2^m - 1 \pmod{2^m}$. This establishes that $U_m \cap \{1, 2, \dots, M\}$ has odd number of elements (so $2^m \mid M+1$) and using $G_{x_m} \in \{2^m - 2, 2^m - 1, 2^m\}$, we iteratively have $U_{m+1} = \{Mu + 2^m c, c = 1, 2, \dots, \lfloor \frac{M}{2^m} \rfloor\}$ whenever $m < k$. $\square$

Now to finish, we note that when $M = k$ we have $U_k = S_k$ so by the previous lemma, $U_k \cap \{1, 2, \dots, M\}$ must have one element. Since $2^k \mid M + 1$ this is only possible when $2^k = M + 1$. With this, all the S_{x_i} 's are uniquely determined and we may show that they map exactly to the number $x_m = \frac{2^{k-m}}{2^k - 1}$.

Combinatorics

- C1. A line in the plane is called *sunny* if it is not parallel to any of the x -axis, the y -axis, or the line $x + y = 0$.

Let $n \geq 3$ be a given integer. Determine all nonnegative integers k such that there exist n distinct lines in the plane satisfying both of the following:

- for all positive integers a and b with $a + b \leq n + 1$, the point (a, b) lies on at least one of the lines; and
- exactly k of the n lines are sunny.

Answer. $k = 0, 1, 3$.

Solution. Let T_n be the set of allowed k 's. The main claim is the following.

T_n is independent of n for all $n \geq 3$. To see why, we easily have $T_n \subseteq T_{n+1}$: just add a line (non-sunny) on the line $x + y = n$. To show $T_{n+1} \subseteq T_n$: consider the outermost triangle: union of points $x + y = n, x = 0$ and $y = 0$. We claim that one of the lines must coincide with one of the three sides, which reduces to the case of n lines. Suppose otherwise, then:

- Each line not under this category has at most two intersects with the three sides in total (e.g. Menelaus') theorem. covers at most 2 lines in the outer triangle. Jointly, this gives at most $2(n + 1)$ intersections.
- On the other hand, there are exactly $3n$ points on the outer triangle, and $3n > 2(n + 1)$.

Thus it suffices to consider $n = 3$. We do three cases here.

$k = 0$ or 1 construction. $x = 0, x = 1$, and any line through $(2, 0)$, which we could choose to be sunny or not.

$k = 3$ construction. $\{(0, 2), (1, 0)\}, \{(2, 0), (0, 1)\}, \{(0, 0), (1, 1)\}$.

Why $k = 2$ fails. We split by cases: if one of the three lines are $x = 0, y = 0$ or $x + y = 2$ then one of other two lines (covering the smaller triangle) must be non-sunny (to cover 2 of out the 3 points). Otherwise, no line can cover more than 2 points (hence each line covers exactly two points). Can verify that the three lines of the $k = 3$ case are the only sunny lines through exactly two points, hence choosing 2 of them means we also need the third.

- C2. There is a row of n paddocks, labelled from left to right with the integers 1 to n , where $n \geq 3$. Skippy the Kangaroo grazes in the paddock according to the following rule:

If Skippy is grazing in paddock k , then she makes a sequence of k hops from a paddock to an adjacent paddock. The first of the k hops is always to the right, unless Skippy is in paddock n , in which case it is to the left. Each of the following $k - 1$ hops is in the same direction as the previous hop, unless Skippy is in paddock 1 or n . Skippy grazes in the paddock that she is in after the k^{th} hop.

For example, if $n = 8$ and Skippy is grazing in paddock 3, the next 4 paddocks she will graze in are 6, 4, 8, 2, in this order. Skippy continues grazing in this way indefinitely. A

paddock is overgrazed if Skippy eventually grazes in it, regardless of which paddock she starts in. Determine all $n \geq 3$ for which there exists exactly one overgrazed paddock.

Answer. $n = 3 \cdot 2^k$.

Solution. Denote $f(k)$ as the next grazing grid of Skippy when she is grazing on k . If k is overgrazed, so is $f(k), f(f(k)), \dots$. Therefore if k is unique overgrazer, $k = f(k)$. We now note the following rule:

$$f(k) = \begin{cases} 2k & k \leq \frac{n}{2} \\ 2(n-k) & \frac{n}{2} < k \leq n-1 \\ 2 & k = n \end{cases}$$

We start with these observations:

- If $k = f(k)$, then $k = \frac{2n}{3}$, or $3 \mid n$.
- If $k \neq n$, and $\nu_2(k) \leq \nu_2(n)$, then $\nu_2(f(k)) > \nu_2(k)$.
- If $k \neq n$, $p > 1$ odd, $p \mid n$, then $p \mid k$ if and only if $p \mid f(k)$.

In particular, the first bullet point allows us to disregard powers of 2. The next we will eliminate is where n is divisible by pq where $p, q > 1$ are odd. The last bullet point suggests that Skippy will only cycle among grids not divisible by p when she starts at 1. However, if she starts at p , the subsequent grids she stays at cannot be divisible by pq , and therefore cannot land at n at all. On the other hand, this also means all the grids it stays are divisible by p . Thus the two paths has two terminal cycles that are disjoint, showing that there is no overgrazed pollock for tis n .

We are left with the case $n = p \cdot 2^k$, where p is prime. Given the requirement $3 \mid n$, we have $n = 3 \cdot 2^k$ for $k \geq 0$. We now claim 2^{k+1} is the only overgrazed pollock: in fact, all paths would terminate here. When $m = 2^{k+1}$, $f(m) = m$. When $m = n$, $f(m) = 2$ and eventually leads to 2^{k+1} . In all other cases $\nu_2(f(m)) \geq \nu_2(m) + 1$, so ν_2 increases as Skippy goes (possibly hitting $m = n$ but then goes back to the cycle to 2^{k+1} .)

- C3. There are 2025 white balls and 2025 black balls arranged in a row. A balanced segment is a contiguous nonempty segment of balls that contains the same number of white balls as black balls. A balanced removal is the operation of selecting a balanced segment S , removing all the balls in S , and shifting left the balls that were to the right of S so that the remaining balls form a contiguous row.

Determine the smallest positive integer N satisfying the following property: for every configuration of balls and for every integer k satisfying $0 \leq k \leq 2025$, there exists a sequence of at most N balanced removals after which there are exactly $2k$ remaining balls.

Answer. $N = 2$.

Solution. We generalize to n white and n black balls, $n \geq 4$. To see why $N = 1$ fails, consider any configuration when first 2 and last 2 balls are all white; then $k = 1$ fails here.

To see why $N = 2$ always works, we consider the notion of ‘prefix’ as the subset of first m balls for any m ; call this prefix strict if $m < 2k$; denote suffix similarly (as consecutive m balls). We start with the following observation:

$N = 1$ works when any strict prefix has more white than black balls (or vice versa). W.l.o.g. we only consider the first case, and the task of removing $2(n-k)$ consecutive balls. Let m_g be the difference between white and black balls among balls $g+1, \dots, g+2(n-k)$ for $g = 1, 2, \dots, 2k$. Then $m_{2k} < 0 < m_0$, m_i ’s are all even, and moreover $|m_g - m_{g+1}| \in \{0, 2\}$. Hence the conclusion follows by discrete continuous value theorem.

To finish, consider the smallest prefix P of at least $2k$ balls and is balanced; the first move would be remove the suffix of the whole sequence, with P left. Let Q be the smallest suffix of P that is balanced, and let $m = |P| - 2k > |Q|$ be the number of balls we need to remove. By minimality, the condition of our previous claim applies, hence we may remove m balls from this suffix Q in one step.

- C4. Bluey is placing lollies on a 1000×1000 grid. Initially, every cell of the grid is empty. At each step, Bluey chooses a row or column in which the total number of lollies is a multiple of 3, chooses an empty cell in this row or column, and places a lolly in it. If there is no such row or column, Bluey stops.

Determine the maximum number of lollies that Bluey can place.

Answer. 3995 (or $4n - 5$ for all $n \geq 5$).

Solution. *Bound:* let the ‘score’ of each row and column be 0, 1, 2 when the number of lollies is congruent to 0, 2, 1, respectively. Denote total score as total row score and total column score across all n rows and columns (i.e. range 0 to $4n$). Consider each of the lolly operation (modulo 3) :

$$(0, 0) \rightarrow (1, 1), (0, 1) \rightarrow (1, 2), (0, 2) \rightarrow (1, 0)$$

Call this type 0, 1, 2. Corresponding score update is $+4, +1, +1$, respectively, and the first update must be type 0 (hence $+4$ points). Now the last update:

- If it's type 0, max total number of steps is $4n - 8 + 2 = 4n - 6$;
- If type 1 or 2, then there is row (and column) where the score is ≤ 2 . The total row score and total column score must align modulo 3 (hence $\leq 2n - 1$ each); total score $\leq 4n - 2$ so including the first 4-point update total number of steps is $\leq 4n - 5$.

Construction: first do

$$(1, 1), (i, i + 1), (i + 1, i), i = 1, 2, \dots, n - 2$$

Now consider permutation π such that $\pi(1) = n$, $\pi(n) = n - 1$ and $|\pi(i) - \pi(i + 1)| \neq 1$ for all $i \geq 1$. E.g. for n even we may do

$$1, 3, \dots, n - 3, n, n - 2, \dots, 2, n - 1$$

for n odd:

$$1, 3, \dots, n, 2, 4, \dots, n - 1$$

Then row/column $\pi(1)$ has 0 lolly, $\pi(n)$ has 1, and $\pi(i)$ has 2 for $i = 2, \dots, n - 1$. Then do

$$(\pi(i), \pi(i + 1)), (\pi(i + 1), \pi(i)), i = 1, 2, \dots, n - 1$$

which gives total of $4n - 5$ operations.

- C5. Let $a_1, a_2, \dots, a_n$ be a permutation of $1, 2, \dots, n$. A pair (i, j) with $i < j$ is called an inversion if $a_i > a_j$. An inversion is called odd if $i - j$ is odd.

- Prove that the total number of inversions is at most 5 times the number of odd inversions.
- There is a permutation such that the total number of inversions is more than 4.99 times the number of odd inversions.

Solution.

(a) Define ‘even inversion’ similarly. Note that for any inversion (i, j) with $i < j$, any k with $i < k < j$, one (or both) of (i, k) and (k, j) must also be an inversion.

We construct a bipartite graph with one side even inversion, and the other side odd inversion, in the following manner: for every even inversion (i, j) with $j - i = 2d > 0$, if d is odd, we connect an edge of weight 1 to either $(i, i + d)$ or $(i + d, j)$, whichever is an inversion; if d is even, we consider one of $(i, i + d - 1), (i + d - 1, j)$ that is an inversion, and also one of $(i, i + d + 1), (i + d + 1, j)$ that is an inversion, and connect (i, j) to each of the chosen inversions with weight $\frac{1}{2}$. Now note that each even inversion is incident edges of total weight 1, thus the total weight of edges is $A - B$. Meanwhile for each odd inversion (i, j) with $j = i + d$, the possible neighbours are:

$$(i, j + d), (i - d, j), (i, j + d + 1), (i - d - 1, j), (i, j + d - 1), (i - d + 1, j)$$

The first two (if chosen) have weights 1 each and the remaining four have weight $\frac{1}{2}$ each, so each of odd inversions are incident to edges of total weight at most $2 + 4 \cdot \frac{1}{2} = 4$. It follows that we have $A - B \leq 4B$ or $A \leq 5B$.

(b) Let $k, m > 1$ and $n = 2km$. We consider m blocks, each $2k$ elements. Then organize the n numbers as follows:

- We consider $2m$ classes of k numbers each, if $i < j$ then elements in class i is strictly smaller than elements in class j (w.l.o.g. can treat the k elements of class j as $jk, jk + 1, \dots, (j + 1)k - 1$).
- Each block gets two classes; the only inversion at block level is that for two adjacent blocks i and $i + 1$, bigger class at block i is bigger than smaller class at block $i + 1$.
- Within each block i of classes a, b ($a < b$), the numbers are presented in alternating and decreasing order; if i is odd then odd indices within this block take class a and even indices take class b . Else the opposite.

E.g. when $k = m = 4$, the classes will take the form

$$(0, 2), (1, 4), (3, 6), (5, 7)$$

The first block will have number (denoted in (class, number) for brevity so (a, b) means $ak + b$)

$$(0, 3), (2, 3), (0, 2), (2, 2), (0, 1), (2, 1), (0, 0), (2, 0)$$

and the second block

$$(4, 3), (1, 3), (4, 2), (1, 2), (4, 1), (1, 1), (4, 0), (1, 0)$$

We now claim that when m is even, $B = \frac{k^2m}{2}$, and $A - B = k(k - 1)m + k^2(m - 1)$.

Now, inversion involving pair of different blocks could only happen for adjacent blocks: bigger class at block i vs smaller class at block $i + 1$ ($i = 1, 2, \dots, m - 1$); by construction, such inversions must be even and contributes to k^2 each, thus contributing $k^2(m - 1)$ to $A - B$.

Within the same block, the two sets of k numbers of the same class are of even inversion, thereby contributing to $2\binom{k}{2} = k(k - 1)$ inversions. Taken together, this contributes another $k(k - 1)m$ to $A - B$. Finally, for the same block i but different classes, around half are odd inversions: $\binom{k}{2}$ for i odd but $\binom{k+1}{2}$ for i even. This gives a total of $\frac{k^2m}{2}$ odd inversions. Therefore we have

$$\frac{A}{B} = 1 + \frac{k(k - 1)m + k^2(m - 1)}{k^2m/2} = 1 + \frac{2(k - 1)}{k} + \frac{2(m - 1)}{m} = 5 - \frac{2}{k} - \frac{2}{m}$$

Taking $k = m = 1000$ solves the problem.

C7. Let P be a regular 100-gon. An Aussie triangulation is formed by dividing P into a set T of non-overlapping triangles such that:

- there are exactly 2025 different points, including the 100 vertices of P that are a vertex of at least one triangle in T , and
- no three of these 2025 points are collinear.

A quokkalateral is a convex quadrilateral that consists of two triangles in T sharing a common side. Two quokkalaterals may share a triangle in T . Determine the minimum number of quokkalateral among all Aussie triangulation.

Answer. 1011.

Solution.

Replacing 2025 and 100 with n and m respectively, with $n > 2(m-1)$ and n odd, we show that in the configuration with n points where the convex hull has exactly m points has a minimum of ? quokkalaterals. Call the $n - m$ points inside the convex hull the 'internal' points, and 'internal' lines as all lines forming the triangle other than the m sides of the convex hull. Denote the *candidate quadrilaterals* as quadrilaterals that has two triangles sharing a common side; a quokkalateral is a candidate quadrilateral.

To begin, consider following setup:

- *Number of triangles.* Each internal point contributes total angle of 2π to the triangles. Together with the $(m-2)\pi$ contributed by the convex hull, the number of triangles is $(m-2) + 2(n-m) = 2n - m - 2$.
- *Number of internal lines.* Each convex hull side and internal line corresponds to 1 and 2 triangles, respectively. Consequently, the number of internal lines are $\frac{3(2n-m-2)-m}{2} = 3n - 2m - 3$.
- *Line-to-quadrilateral bijection.* Each candidate quadrilateral corresponds to exactly one internal line (i.e. there is a bijection between the two).
- *Counting and minimizing quokkalateral.* Consider any candidate quadrilateral; we need to check the internal lines in T correspond to convex angles to determine if it is quokkalateral. In other words, minimizing quokkalaterals (among all the allowable configurations) is equivalent to maximizing internal lines in T that has exactly one endpoint where the triangles adjacent to it forms a reflex angle.

In essence, we can also count how many reflex angles are contributed by each point in T ; such points have to be internal (not the m points on the convex hull).

Before proceeding, we define a *supertriangle* as a collection of 3 triangles OAB, OAC, OBC such that O is inside triangle ABC ; call O its center for this purpose. The main claim is the following.

Lemma 5. *An internal point contributes to exactly 3 reflex angles if and only if it is the center of a supertriangle. Moreover, an internal point contributing to exactly two reflex angles must correspond to a vertex of a triangle in T that cannot be part of a supertriangle; each such internal point must correspond to different triangle in T .*

Proof. For the first part, the 'if' part is clear. To solve for 'only if' part, if a point O corresponds to at least two reflex angles, these two reflex angles must overlap, hence share a triangle in T . The only possibility when each of the 3 pairs of reflex angles overlap is when this point O is incident to exactly three internal lines, which then forms a supertriangle.

For the second part, consider the angle shared between two reflex angles incident to O . This angle corresponds to triangle in T where O is a vertex. Now, O cannot be the center

of a supertriangle. Each the two regions in O adjacent to this angle add up to a reflex angle with this region, hence O cannot be part of a supertriangle either. Finally, suppose this triangle also corresponds to another point R with two reflex angles. This point R must also be a vertex of this triangle and the line OR has both endpoints (O and R) corresponding to a reflex angle. This is impossible as no quadrilaterals can have two reflex angles. $\square$

Now, let k_3 be the number of supertriangles, k_2 number of points contributing to two reflex angles, and k_1 the number of remaining points. Then $k_1 + k_2 + k_3 = n - m$ and:

$$k_3 \leq \min\left\{\frac{2n - m - 2}{3}, n - m\right\} \quad k_2 \leq \min\{2n - m - 2 - 3k_3, n - m - k_3\}$$

where $2n - m - 2$ denotes the total number of triangles, and $2n - m - 2 - 3k_3$ denotes the non super triangles. Then the number of nonconvexity is upper bounded by

$$3k_3 + 2k_2 + k_1 = 2k_3 + k_2 + (n - m)$$

We note that the function $2x + \min\{2n - m - 2 - 3x, n - m - x\}$ is piecewise linear in x that attains maximum value when $2n - m - 2 - 3x = n - m - x$, i.e. when $x = \frac{n-2}{2}$. Since $n \geq 2m - 2$, $\frac{n-2}{2} \leq \min\{\frac{2n-m-2}{3}, n - m\}$. However, our n is odd, hence we need to consider the adjacent values $\frac{n-3}{2}$ and $\frac{n-1}{2}$; both give the value $\frac{3(n-1)}{2} - m$. Thus the number of quokkalateral is at least

$$3n - 2m - 3 - \left(\frac{3(n-1)}{2} - m\right) - (n - m) = \frac{n-3}{2}$$

Construction: (another time, lol)

Geometry

- G1. Let ABC be a triangle such that $\angle A$ is obtuse. Let D and E be points on sides AB and AC , respectively, such that $BDEC$ is cyclic. Suppose the circumcircle of triangle ADE intersects side BC at two points X and Y , where B, X, Y and C lie in that order. Let the circumcircles of triangles BDX and CEY be Ω and Γ , respectively. Let P be a point such that PD is tangent to Ω and PE is tangent to Γ . Let Q be a point such that QB is tangent to Ω and QC is tangent to Γ . Prove that points A, P and Q are collinear.

Solution. We show that A, P, Q all lie on radical axis of Ω and Γ .

A : power to Ω and Γ are $AD \cdot AB$ and $AE \cdot AC$, respectively. The two quantities are equal because B, D, C, E cyclic.

Q : we claim that $BQ = QC$. Note that if O is circumcenter of ADE then $\angle DAO = 90^\circ - \angle AED = 90^\circ - \angle ABC$ so $AO \perp BC$. Thus $AX = AY$ and $\angle CBQ = \angle BDX = \angle CEY = \angle QCB$.

P : similarly, we show $PD = PE$. One sees that $\angle AED = \angle ABC = \angle XDP$ and $\angle ADE = \angle ACB = \angle PEY$, and $\angle BDX = \angle CEY$ as previously claimed.

- G2. Let ABC be an acute-angled triangle. Let I_B and I_C be the excenters of triangle ABC opposite B and C , respectively. Let E be a point on line BA such that $\angle EI_B B = 90^\circ$. Let F be a point on line CA such that $\angle CI_C F = 90^\circ$. Let the circle with center I_B and radius $I_B E$ intersect the segment $I_B B$ at X . Let the circle with center I_C and radius $I_C F$ intersect the segment $I_C C$ at Y . Prove that BY is perpendicular to CX .

Solution. The main claim is that triangles $I_C B Y$ and $I_B X C$ are similar, and that both angles $\angle BI_C Y = \angle CI_B X = \frac{\angle BAC}{2}$. Assuming the claim, we then have

$$\angle YBC + \angle XCB = \angle I_C B C + \angle I_B C B - \angle I_C B Y - \angle I_B C X$$

$$= (90^\circ + \frac{\angle ABC}{2}) + (90^\circ + \frac{\angle ACB}{2}) - (180^\circ - \frac{\angle BAC}{2}) = 90^\circ$$

where the last equality follows from that the sum of 3 angles in a triangle is 180° .

To prove the claim, the angle claim $\angle BI_C Y = \angle CI_B X = \frac{\angle BAC}{2}$ follows directly from angle chasing. In particular, $\angle BI_C Y = 180^\circ - \frac{\angle ACB}{2} - (90^\circ + \frac{\angle ABC}{2}) = \frac{\angle BAC}{2}$. Thus, proving similarity of the triangle now reduces to proving that $\frac{I_C B}{I_C Y} = \frac{I_B X}{I_B C}$. Now $I_C Y = I_C F = I_C C \tan \frac{\angle ACB}{2}$; by sine rule we have $I_C B = I_C C \frac{\sin \frac{\angle BCA}{2}}{\cos \frac{\angle ABC}{2}}$. Therefore $\frac{I_C B}{I_C Y} = \frac{\cos \frac{\angle ACB}{2}}{\cos \frac{\angle ABC}{2}}$. The other equality $\frac{I_B X}{I_B C} = \frac{\cos \frac{\angle ACB}{2}}{\cos \frac{\angle ABC}{2}}$ can be established similarly.

- G3. Let ABC be an acute triangle with $AB < AC$. Let Γ be the circumcircle of ABC . The perpendicular from B to AC intersects Γ again at $D \neq B$. Let Q be the point on Γ , different from B , such that $AB = AQ$. Lines BQ and AC intersect at R . Line DR intersects Γ again at $S \neq D$. Segment BC intersects the circumcircle of triangle QRS at T . Lines TQ and RS intersect at X . Prove that CX is perpendicular to AB .

Solution. We first show that $QT \perp AC$. Let T' on BC such that $QT' \perp AC$. Since $AB = AQ$, $\angle ACB = \angle ACQ$, so QT' is perpendicular bisector of AC . This then gives the following sequence of equalities:

$$\angle RT'Q = \angle RQT' = \angle DBQ = \angle DSQ$$

The second equality due to $BD \parallel QT'$ and last equality due to D, B, S, Q concyclic on Γ . Therefore, T' is on the circle QRS and on segment BC . Given that the other intersection of circle QRS and line BC will be on its extension beyond C , we have $T = T'$.

Next, let AC intersect BD at U , and QT intersect AC at V . Then given $BD \parallel QT$ and D, R, X collinear, $\frac{BU}{UD} = \frac{VQ}{VX}$. This then gives $\frac{\tan \angle CAB}{\tan \angle CAD} = \frac{\tan \angle ACQ}{\tan \angle ACX}$, and given that $\angle CAD = 90^\circ - \angle ACB = 90^\circ - \angle ACQ$, the other two angles $\angle CAB$ and $\angle ACX$ also add up to 90° . This give $CX \perp AB$, as required.

- G4. (IMO 2) Let Ω and Γ be circles with centres M and N , respectively, such that the radius of Ω is less than the radius of Γ . Suppose Ω and Γ intersect at two distinct points A and B . Line MN intersects Ω at C and Γ at D , so that C, M, N, D lie on MN in that order. Let P be the circumcentre of triangle ACD . Line AP meets Ω again at $E \neq A$ and meets Γ again at $F \neq A$. Let H be the orthocentre of triangle PMN .

Prove that the line through H parallel to AP is tangent to the circumcircle of triangle BEF .

Solution. Claim: P is A -excenter of triangle AMN .

Proof: $CM = MA$ and $PC = PA$ so MP is perpendicular bisector of AC (so MP bisects $\angle CMA$), same for NP and $\angle DNA$.

Introduce E_1 midpoint of AE , F_1 midpoint of AF , and H midpoint of AB . Let circle HE_1F_1 intersect CD again at G , and I incneter of triangle AMN . Then:

- H is foot of perpendicular from A to CD ;
- $CE_1 \perp AP$ and $DF_1 \perp AP$;
- HE_1AC and HF_1AD cyclic;
- $\angle E_1HG = \angle CAP = \angle DAP = \angle F_1HG$, so $E_1G = F_1G$
- $\frac{E_1G}{CG} = \frac{\sin \angle E_1CD}{\sin \angle CE_1G}$ and $\frac{F_1G}{DG} = \frac{\sin \angle F_1DC}{\sin \angle DF_1G}$. Note $\angle CE_1G + \angle DF_1G = 180^\circ$ and $\angle E_1CD = \angle F_1DC$, and $E_1G = F_1G$ so G is midpoint of CD .
- Finally, $IMNH$ is parallelogram.

Thus to finish:

- Tangent at G to circle HE_1F_1 is parallel to AP , call this line ℓ_0 .
- Since G is also midpoint of IH , line ℓ_1 through H parallel to AP is $2\times$ distance to AP as compared to line ℓ_0 .
- BEF is $2\times$ homothety compared to HE_1F_1 with center A , so ℓ_1 is tangent to circle BEF .

G5. Let $ABCD$ be a convex quadrilateral with $\angle ADC > 90^\circ$. Suppose that points X and Y lie on diagonal AC such that

$$\angle ADX = \angle YDC = 90^\circ, \quad \angle CAD = \angle CBX, \quad \text{and} \quad \angle DCA = \angle YBA.$$

Let O be the circumcentre of triangle BXY . Suppose point $R \neq O$ lies on the perpendicular bisector of AC such that lines OR and AC are parallel.

Prove that line RO bisects angle $\angle DRB$.

Solution. Let the circumcircle of DAC intersect DY and DX again at P, Q , respectively. Then $APQC$ is a rectangle, and in addition $\angle YBA = \angle DCA = \angle DPA$, so $AYBP$ is cyclic and similarly $CXBQ$ is cyclic. Consequently, $\angle YBP = \angle YAP = 90^\circ$ and similarly $\angle XBQ = \angle XCQ = 90^\circ$.

We now claim that D, O, B are collinear. In fact, let E be such that $\angle EYB = \angle EXB = 90^\circ$, then O is the midpoint of EB . Meanwhile, $EY \parallel BP$ and $EX \parallel BQ$, so D, E, B are also collinear, and thus both D and O are on the line EB .

Denote S as the point on line DB such that $PS = SQ$; this is obtained by sending $SP \parallel YO$ and $SQ \parallel OX$. Observe that S and R are both on the perpendicular bisector of AC . We now claim that $(D, B; O, S)$ is a harmonic bundle, i.e. $\frac{OB}{OD} = \frac{SB}{SD}$. Now, the homothety map Γ centered at D that brings Y to P has $\Gamma(X) = Q$, $\Gamma(E) = B$ and $\Gamma(O) = S$. Therefore, $\frac{SB}{SD} = \frac{OE}{OD}$. Given that $OE = OB$ as O is midpoint of EB , this establishes our claim.

Finally, since $OR \perp RS$ and $(D, B; O, S)$ harmonic, it follows that OR and RS are internal and external bisectors of $\angle DRB$, respectively.

Number Theory

N1. Let n and k be positive integers such that $n < k < 2n$. Suppose $a_1, a_2, \dots, a_k$ are positive integers such that $2^{a_1} + 2^{a_2} + \dots + 2^{a_k}$ is divisible by $2^n - 1$.

Show that at least three of $a_1, a_2, \dots, a_k$ have the same remainder when divided by n .

Solution. Take all exponents modulo n . Let $a = \sum 2^{b_j}$ and $b = \sum 2^{c_j}$, b_j are different indices occurring exactly twice and c_j exactly once. Then $2^n - 1 \mid 2a + b$, and a, b have disjoint positions of 1s (binary form). a, b cannot be 0 given $n < k < 2n$.

Case b odd: $2a + b = 2^n - 1$ and $2a, b$ have complementary digits at each place: rightmost 0 of $b = 1$ for $2a$, its right is 1 for both a and b , contradiction.

Case b even: $2a + b = 2(2^n - 1)$. a and $\frac{b}{2}$ have complementary digits at each place: leftmost 1 of $\frac{b}{2}$: to the left is 0 for $\frac{b}{2}$ but 1 for both a and b , contradiction.

N2. In an $n \times n$ board, for each $1 \leq i \leq n$ and $1 \leq j \leq n$, the cell in the i th row from the bottom and j th column from the left contains $\gcd(i, j)$ leaves. A koala travels from the bottom left corner to the top right corner. At each step, the koala moves up or to the right by a single cell. The koala eats the leaves in each cell it visits, including the bottom left and top right cells.

Determine the maximum number of leaves that the koala can eat.

Answer. $\frac{n(n+1)}{2} + (n-1)$.

Solution. *Construction.* This path is in zigzag manner (i.e. $(k-1, k)$ on odd-numbered step and (k, k) on even-numbered step).

Bound. Let the initial step index be 2 and ending step index be $2n$. start at step 2 (w.l.o.g.) and end at step $2n$; the sum of coordinates at step indexed i is i . We show that the total number of leaves at steps $2k-1$ and $2k$ cannot exceed $k+1$. To proceed, one of the grids must have strictly more than $\frac{k}{2}$ leaves. This means we are down to one of the following 3 cases.

Step $2k$ contains k leaves. Then the coordinate at step $2k$ is (k, k) , at step $2k-1$ is either $(k-1, k)$ or $(k, k-1)$ and can only contain 1 leaf.

Step $2k$ contains $\frac{2k}{3}$ leaves. By symmetry, we may assume that the coordinate is $(a, b) = (\frac{2k}{3}, \frac{4k}{3})$. Crucially, $b = 2a$. Then the coordinate at step $2k-1$ is either $(a-1, 2a)$ or $(a, 2a+1)$, which has $\gcd \leq 2$. Hence the total number of leaves is at most $2 + \frac{2k}{3} \leq k+1$, since $k \geq 3$.

Step $2k-1$ contains $\frac{2k-1}{3}$ leaves. We can similarly deduce that the coordinates must be (a', b') with $b' = 2a'$, and that the coordinates at step $2k$ is either $(a', 2a'+1)$ or $(a'+1, 2a')$. Either way, the $\gcd$ is at most 2 and total number of leaves is $\leq 2 + \frac{2k-1}{3} \leq k+1$ for $k \geq 2$.

N3. A proper divisor of a positive integer N is a positive divisor of N other than N itself.

The infinite sequence $a_1, a_2, \dots$ consists of positive integers, each of which has at least three proper divisors. For each $n \geq 1$, the integer a_{n+1} is the sum of the three largest proper divisors of a_n .

Determine all possible values of a_1 .

Answer. $a_1 = 12^k \cdot 6c$, c relatively prime to 2 and 5.

Solution. The main insight is that at no point can a_n be monotonically strictly decreasing. This gives the following two claims.

a_n must be even. Otherwise, $a_{n+1} \leq a_n(\frac{1}{3} + \frac{1}{5} + \frac{1}{7}) < a_n$ and a_{n+1} is odd, so a_m is monotonically decreasing for all $m \geq n$.

a_n must be divisible by 3. Suppose otherwise. Then $a_{n+1} = a_n(\frac{1}{a} + \frac{1}{b} + \frac{1}{c})$ where $a < b < c$ are the smallest divisors less than n , and greater than 1. Note that $a \geq 2, b \geq 4, c \geq 5$, so $a_{n+1} \leq \frac{19}{20}a_n < a_n$. Moreover, we assumed a_n is even, so $a = 2$. If $b = 4, 3 \mid a_n(\frac{1}{a} + \frac{1}{b})$ so $3 \nmid a_{n+1}$. If $b > 4, 4 \nmid a_n$, b the smallest odd divisor of a_n (i.e. odd prime). Then $c = 2b$. Now can show $a_{n+1} = a_n \cdot \frac{b+3}{2b}$ which is not divisible by 3 (as $3 \nmid b$), so this cycle continues and a_m is monotonically decreasing for all $m \geq n$.

Now with these set up, we now have the three smallest divisors greater than 1 as $2, 3, c$ where $c \in \{4, 5, 6\}$. Note, however, the case $c = 5$ can't occur as this means a_n is divisible by 2 but not 4, and so $a_{n+1} = \frac{31}{30}a_n$ is odd, which has been shown to be impossible. Call a number in this form type- c ; a_n of type 4, 6 gives $a_{n+1} = \frac{13}{12}a_n, a_n$ respectively. Since 13 and 31 are relatively prime to 4, 5, 6, so a_{n+1} can never become a smaller type than a_n . It follows that a_1 has form $12^k A$ where $6 \mid A$ but $4, 5 \nmid A$. The terminating number in the sequence is then $13^k A$.

N4. Find all functions $f : \mathbb{N} \rightarrow \mathbb{Z}$ such that for every positive integers n and d such that $d \mid n$, there exists a positive integer $e \mid n$ such that $n \mid d + f(e)$.

Answer. $f(x) = -x$. This fulfills the condition by taking $e = d$.

Solution. The equivalent statement of the problem condition is for each n , $\{f(d) : d \mid n\} = \{-d : d \mid n\} \pmod{n}$. Denote this condition as $(*)$. In particular, $(*)$ implies injectivity since $f(d_1) = f(d_2)$ implies $d_1 = d_2$ when taking $n = kd_1d_2$ for $k = (|f(d_1)| + |f(d_2)| + 1)$. The main claim is the following.

Lemma 6. For each $d > 0$, $f(d) < 0$ and $f(d) \mid m$.

Proof. For each $d, k \geq 1$ we choose e_k as the number where $dk \mid f(d) + e_k$ and $e_k \mid dk$. Now choose k where $dk > 2|f(d)|$. Note that either $\frac{1}{dk/e} = 1$ or $\leq \frac{1}{2}$, the first case implies $f(d) = 0$; the second case implies $\frac{f(d)}{dk} = -\frac{1}{k'} < 0$ for some k' depending on k . Thus $f(d) \mid dk$ for all k sufficiently large, so $f(d) \mid d$,

We're now left with showing $f(d)$ cannot be 0. By injectivity, there is at most one such d_0 with $f(d_0) = 0$. If this d_0 exists, then this sequence of logic holds:

- By injectivity, and $f(d) \mid n$ for $d \mid n$ but $d_0 \nmid n$, we have $f(n) = -n$ for n with $d_0 \nmid n$.
- By injectivity again we have $d_0 \mid f(n)$ whenever $d_0 \nmid n$.
- Consider $n = 2d_0, 3d_0$ simultaneously, however, $f(n) = -kd_0$. Given also that $f(d_0) = 0$, we have $f(2d_0) = f(3d_0) = -d_0$, contradicting injectivity.

□

To see why this claim uniquely determines f , we construct each $f(1), f(2), \dots$ as follows: for each n , assume that $f(d) = -d$ for all $d = 1, 2, \dots, n-1$, the injectivity claim implies that the only number smaller than $-(n-1)$ and is a divisor of n is $-n$.

- N5. The sequence of triples of positive integers $(a_0, b_0, c_0), (a_1, b_1, c_1), \dots, (a_{2025}, b_{2025}, c_{2025})$ satisfies

$$(a_{i+1}, b_{i+1}, c_{i+1}) = (a_i + \gcd(b_i, c_i), b_i + \gcd(c_i, a_i), c_i + \gcd(a_i, b_i))$$

for $0 \leq i \leq 2024$. Determine the smallest possible value of a_{2025} .

Answer. $3 \cdot 2^{675}$.

Solution. *Construction:* take $(a_0, b_0, c_0) = (3, 5, 6)$, then

$$(3, 5, 6) \rightarrow (4, 8, 7) \rightarrow (5, 9, 11) \rightarrow (6, 10, 12)$$

i.e. doubling every 3 sequences.

Bound: note the map $(a_n, b_n, c_n) \rightarrow (a_{n+1}, b_{n+1}, c_{n+1})$ is permutation equivariant and scale preserving (i.e. if $(ka_n, kb_n, kc_n) \rightarrow (ka_{n+1}, kb_{n+1}, kc_{n+1})$). In modulo 2, we have the following iterations:

$$(0, 1, 1) \rightarrow (1, 0, 0) \rightarrow (1, 1, 1) \rightarrow (0, 0, 0)$$

Each triple with (a, b, c) not all even falls into one of the first three (up to permutations), i.e. $g(a, b, c) := \min\{\nu_2(a), \nu_2(b), \nu_2(c)\}$ (the highest power of 2 dividing all entries) increases in at most 3 iterations. Thus for $i = 0, 1, 2$ we have $a_{3k+(i+1)} - a_{3k+i} \geq 2^k$ as $g(a_{3k+i}, b_{3k+i}, c_{3k+i}) \geq k$. Thus we have

$$a_{2025} - a_0 \geq \sum_{i=0}^{674} (2^i + 2^i + 2^i) = 3(2^{675} - 1)$$

and given also $2^{675} \mid a_{2025}$ we have $a_{2025} \geq 2^{675}$.

- N6. For positive integers n and k , let $f_k(n)$ denote the remainder when n is divided by $2^k - 1$. A positive integer n is grouseif

$$f_1(n) \leq f_2(n) \leq f_3(n) \leq f_4(n) \leq \dots$$

- (a) Prove that there are infinitely many grouse numbers.

- (b) Prove that there exists a positive integer M such that there are at most $\frac{M}{2025^{2025}}$ grouse numbers less than M .

Solution. (a) For each $k \geq 2$, consider the following number obtained by the following protocol:

$$s_0 = 2^k - 1 \quad s_j = \lceil \frac{s_{j-1}}{2^{k-j}-1} \rceil \cdot (2^{k-j} - 1), \forall j = 1, 2, \dots, k-1$$

We claim that s_{k-1} is grouse, and that $2^k - 1 \leq s_{k-1} < 2^{k+1} - 1$. Indeed, we have $0 \leq s_j - s_{j-1} \leq 2^{k-j} - 2$. This means that $0 \leq s_{k-1} - s_j = \sum_{i=1}^{k-j-1} (2^i - 2) = 2^{k-j} - 2(k-j) < 2^{k-j} - 1$. Given also that s_j is divisible by $2^{k-j} - 1$, we have $f_{k-j}(s_{k-1}) = s_{k-1} - s_j$. Consequently, given that $s_0 \leq s_1 \leq \dots \leq s_{k-2} \leq s_{k-1}$, we have $f_1(s_{k-1}) \leq f_2(s_{k-1}) \leq \dots \leq f_{k-1}(s_{k-1}) \leq f_k(s_{k-1})$. Finally, given that $s_{k-1} \leq s_0 + 2^k - 2k \leq 2^{k+1} - 2k < 2^{k+1} - 1$, we have $f_m(s_{k-1}) = s_{k-1}$ for $m \geq k+1$. Thus this s_{k-1} is grouse and there is at least one grouse number between the range $2^k - 1 \leq n < 2^{k+1} - 1$ for all $k \geq 2$.

(b) We show that if $n \geq 2^k - 1$ is grouse, then $n \geq (2^{m_1} - 1)(2^{k+1-m_1} - 1)$ for all integers $m_1 = 1, 2, \dots, k$. A quick observation is the following.

If a number s is grouse and $s \geq c \cdot (2^m - 1)$ then $s \geq t$ with $t = (2^{m-1} - 1) \cdot \lceil c \frac{2^m - 1}{2^{m-1} - 1} \rceil$. Indeed, if $s = c \cdot (2^m - 1) + u$ where $u \geq 0$ then $f_m(s) \leq u$ but if $s < t$, then $f_{m-1}(c \cdot (2^m - 1)) > 0$ and then $f_{m-1}(s) = f_{m-1}(c \cdot (2^m - 1)) + u > u$ so this s is not grouse.

Plugging this back, we may do induction with m_1 (starting from $m_1 = 1$) and the conclusion follows from that $\lceil (2^{m_1} - 1) \frac{2^{k+1-m_1} - 1}{2^{k-m_1-1} - 1} \rceil \geq 1 + 2(2^{m_1} - 1) = 2^{m_1+1} - 1$.

To finish, this means for any m_1 there are at most $2^{m_1} + 2^{k+1-m_1}$ grouse numbers in the range $\{2^k - 1, \dots, 2^{k+1} - 2\}$. Choosing $m_1 = \lceil \frac{k+1}{2} \rceil$, we deduce that there is a constant $c > 0$ such that for any $M > 0$ there are at most $c\sqrt{M}$ grouse numbers that is at most M . This solves the problem.

- N7. Let $\mathbb{N}$ denote the set of positive integers. A function $f: \mathbb{N} \rightarrow \mathbb{N}$ is said to be bonza if

$$f(a) \text{ divides } b^a - f(b)^{f(a)}$$

for all positive integers a and b .

Determine the smallest real constant c such that $f(n) \leq cn$ for all bonza functions f and all positive integers n .

Answer. $c = 4$.

Solution. *Bound.* Set $a = b$, then $f(a) \mid a^a$. Suppose for some prime p we have $p \mid f(a)$ (which then means $p \mid a$). Then take any b divisible by p , $p \mid f(a) \mid b^a - f(b)^{f(a)}$ so $p \mid f(b)$. This means given a prime p , if $p \mid f(a)$ for some a then $p \mid f(a)$ for all a with $p \mid a$. Let S be the set of primes dividing at least one value $f(a)$. Crucially, we note that for $p \in S$, $p \mid f(p) \mid p^p$ for any $p \in S$ so $f(p)$ is a (positive) power of p . Note that $p^k \equiv 1 \pmod{p}$, so $b^{p^k} \equiv b \pmod{p}$ by Fermat's little theorem, giving the following:

$$p \mid f(p) \mid b^p - f(b)^{f(p)} \equiv b - f(b) \pmod{p}$$

for any b , so $p \mid b - f(b)$.

Assume S is infinite. Then for each b , and consider $p \in S$ that's greater than both b and $f(b)$. The fact $p \mid b - f(b)$ forces $f(b) = b$, hence f is an identity function and so $c = 1$ in this case.

In another case, let S be finite. Note that $S = \emptyset$ implies $c = 1$ too since $f \equiv 1$, so assume S is nonempty. Let p_0 be the largest element of S . Then for all primes $q > p_0$ we have $f(q) = 1$ so $p_0 \mid q - 1$ for all such primes q . However, by Dirichlet's theorem,

for any r with $\gcd(r, p_0) = 1$ there are infinitely many primes q such that $p_0 \mid q - r$, so $\gcd(r, p_0) = 1 \rightarrow p_0 \mid r - 1$. This forces $p_0 = 2$. In other words, $f(a)$ is a power of two for all a , with $f(a) = 1$ for all odd numbers a . Now set $b = 3$ and let $a = c \cdot 2^k$ be even, with $c = 1$. Then

$$f(a) \mid 3^a - 1 = (3^c - 1)(3^c + 1) \cdots (3^{2^{k-1}c} + 1)$$

and note that $3^c - 1, 3^{2^c} + 1, \dots, 3^{2^{k-1}c} + 1$ are all divisible by 2 but not by 4, and $3^c + 1$ divisible by 4 but not by 8. Thus $\nu_2(3^a - 1) = k + 2$, i.e. $f(a) \leq 2^{k+2} \leq 4a$. Given also $f(a) = 1$ for all odd a we have $c \leq 4$.

Construction. The example above motivates the following construction:

$$f(a) = \begin{cases} 1 & a \text{ odd} \\ 4 & a = 4k + 2 \\ 16 & 4 \mid a \end{cases}$$

Here $f(4) = 16$ suggests $c = 4$ cannot be improved. To verify that this function works, we need to show that

$$4 \mid b^a - f(b)^4, a = 4k + 2 \quad 16 \mid b^a - f(b)^{16}, a = 4k$$

In the first case we automatically have $4 \mid b^a, f(b)^4$ when a, b both even, and when b is odd we have $b^a \equiv 1 \pmod{4}$ (since b^a is a square). In the second case we have $16 \mid b^a, f(b)^{16}$ when b even and $a \geq 4$, and when b is odd we note that $b^4 - 1 = (b - 1)(b + 1)(b^2 + 1)$ is divisible by 16.

N8. Prove that there are finitely many prime numbers p such that

- $p - 8$ is a perfect square, and
- for all odd positive integers $k < \sqrt{p}$, the number $p - k^2$ has at most two distinct prime factors.

Solution. We consider the following claim.

Claim: all such p 's must be in the form of $4 \cdot 2^k \cdot (2^k \pm 1) + 9$ where the last factor $2^k \pm 1$ is odd prime.

Proof: note that p is odd and $p - 9 = a^2 - 1$ for some a (hence even), so can have at most one odd factor. Given $\gcd(a - 1, a + 1) = 2$, one of them is power of 2 and the other one is $2 \cdot$ odd prime power. But the only two consecutive perfect powers are 8 and 9, hence this 'prime power' must be itself a prime.

Now we may split into the following cases.

Case $2^k + 1$: for this to be prime, k must be power of 2, i.e. $k = 2^m$. Then $p = 4 \cdot 2^k(2^k + 1) + 9 = (2^{k+1} + 1)^2 + 8$. When considering modulo 17, for $k = 2^m$ with $m \geq 3$ we have

$$(2^{2^m+1} + 1)^2 + 8 \equiv (2 + 1)^2 + 8 \equiv 0 \pmod{17}$$

as $2^8 \equiv 1 \pmod{17}$, thus this p is not prime for all such p 's.

Case $2^k - 1$: for this to be prime, k must be itself a prime (in particular, k is odd). We have a similar expansion $p = (2^{k+1} - 1)^2 + 8$.

Claim: when $k \geq 7$, there exists a prime number $q \leq 2p^{1/4}$ such that p is a quadratic residue modulo p .

Proof: we will find those q where $q \mid 2^{k+1} - 1$, and 2 (and therefore 8) is a quadratic residue. When $k + 1$ is not a power of 2, let $m > 1$ be any odd divisor of $k + 1$, then $2^m - 1 \mid 2^{k+1} - 1$. Take any prime $q \mid 2^m - 1$, since m is odd, we have order modulo $\text{ord}_q(2)$ odd as $\text{ord}_q(2) \mid m$. But $\phi(q) = q - 1$ is even, so 2 is indeed a quadratic residue.

When $k+1$ is a power of 2, we have the expansion $2^{k+1} - 1 = (2^{\frac{k+1}{2}} - 1)(2^{\frac{k+1}{2}} + 1)$ and $2^{\frac{k+1}{2}} + 1$. We now use the identity that a) 2 is a quadratic residue for prime q if and only if $q \equiv 1, 7 \pmod{8}$, and b) all prime divisors of $x^{2^m} + 1$ are of the form $\ell \cdot 2^{m+1} + 1$, as $\text{ord}_q(x) = 2^{m+1}$. Thus when $k+1 \geq 8$ (i.e. $m \geq 2$), 2 is a quadratic residue of any such prime q .

Taken together, all such q 's are at most $2^{\frac{k+1}{2}} + 1 \leq 2(2^{\frac{k+1}{2}} - 1) \leq 2\sqrt{2^{k+1} - 1} \leq 2p^{1/4}$, proving our claim.